

INDR 252 Midterm 2 Study Questions

1)

A certain type of stainless steel powder is supposed to have a mean particle diameter of $\mu = 15 \mu\text{m}$. A random sample of 87 particles had a mean diameter of $15.2 \mu\text{m}$, with a standard deviation of $1.8 \mu\text{m}$. A test is made of $H_0 : \mu = 15$ versus $H_1 : \mu \neq 15$.

- Find the P -value.
- Do you believe it is plausible that the mean diameter is $15 \mu\text{m}$, or are you convinced that it differs from $15 \mu\text{m}$? Explain your reasoning.

2)

A new concrete mix is being designed to provide adequate compressive strength for concrete blocks. The specification for a particular application calls for the blocks to have a mean compressive strength μ greater than 1350 kPa. A sample of 100 blocks is produced and tested. Their mean compressive strength is 1356 kPa and their standard deviation is 70 kPa. A test is made of $H_0 : \mu \leq 1350$ versus $H_1 : \mu > 1350$.

- Find the P -value.
- Do you believe it is plausible that the blocks do not meet the specification, or are you convinced that they do? Explain your reasoning.

3)

If $P = 0.50$, which is the best conclusion?

- H_0 is definitely false.
- H_0 is definitely true.
- There is a 50% probability that H_0 is true.
- H_0 is plausible, and H_1 is false.
- Both H_0 and H_1 are plausible.

4)

A machine that fills cereal boxes is supposed to be calibrated so that the mean fill weight is 12 oz. Let μ denote the true mean fill weight. Assume that in a test of the hypotheses $H_0 : \mu = 12$ versus $H_1 : \mu \neq 12$, the P -value is 0.30.

- Should H_0 be rejected on the basis of this test? Explain.
- Can you conclude that the machine is calibrated to provide a mean fill weight of 12 oz? Explain.

5)

A grinding machine will be qualified for a particular task if it can be shown to produce less than 8% defective parts. In a random sample of 300 parts, 12 were defective. On the basis of these data, can the machine be qualified?

6)

A certain manufactured product is supposed to contain 23% potassium by weight. A sample of 10 specimens of this product had an average percentage of 23.2 with a standard deviation of 0.2. If the mean percentage is found to differ from 23, the manufacturing process will be recalibrated.

- a. State the appropriate null and alternate hypotheses.
- b. Compute the P -value.
- c. Should the process be recalibrated? Explain.

7)

In a certain supermarket, a sample of 60 customers who used a self-service checkout lane averaged 5.2 minutes of checkout time, with a standard deviation of 3.1 minutes. A sample of 72 customers who used a cashier averaged 6.1 minutes with a standard deviation of 2.8 minutes.

- a. Can you conclude that the mean checkout time is less for people who use the self-service lane?
- b. Can you conclude that if everyone used the self-service lane, that the mean checkout time would decrease?
Consider the number of items checked out when formulating your answer.

8)

In a study conducted by the U.S. Department of Health and Human Services, a sample of 546 boys aged 6–11 was weighed, and it was determined that 87 of them were overweight. A sample of 508 girls aged 6–11 was also weighed, and 74 of them were overweight. Can you conclude that the proportion of boys who are overweight differs from the proportion of girls who are overweight?

9)

A new post-surgical treatment was compared with a standard treatment. Seven subjects received the new treatment, while seven others (the controls) received the standard treatment. The recovery times, in days, are given below.

Treatment: 12 13 15 19 20 21 24

Control: 18 23 24 30 32 35 39

Can you conclude that the mean recovery time for those receiving the new treatment is less than the mean for those receiving the standard treatment?

10)

The article "Determination of Carboxyhemoglobin Levels and Health Effects on Officers Working at the Istanbul Bosphorus Bridge" (G. Kocasoy and H. Yalin, *Journal of Environmental Science and Health*, 2004:1129–1139) presents assessments of health outcomes of people working in an environment with high levels of carbon monoxide (CO). Following are the numbers of workers reporting various symptoms, categorized by work shift. The numbers were read from a graph.

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	Shift		
	Morning	Evening	Night
Influenza	16	13	18
Headache	24	33	6
Weakness	11	16	5
Shortness of Breath	7	9	9

Can you conclude that the proportions of workers with the various symptoms differ among the shifts?

11)

Scores on the math SAT are normally distributed. A sample of 20 SAT scores had standard deviation $s = 87$. Someone says that the scoring system for the SAT is designed so that the population standard deviation will be $\sigma = 100$. Do these data provide sufficient evidence to contradict this claim?

12)

A broth used to manufacture a pharmaceutical product has its sugar content, in mg/mL, measured several times on each of three successive days.

Day 1: 5.0 4.8 5.1 5.1 4.8 5.1 4.8
4.8 5.0 5.2 4.9 4.9 5.0
Day 2: 5.8 4.7 4.7 4.9 5.1 4.9 5.4
5.3 5.3 4.8 5.7 5.1 5.7
Day 3: 6.3 4.7 5.1 5.9 5.1 5.9 4.7
6.0 5.3 4.9 5.7 5.3 5.6

- Can you conclude that the variability of the process is greater on the second day than on the first day?
- Can you conclude that the variability of the process is greater on the third day than on the second day?

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Scores on a certain IQ test are known to have a mean of 100. A random sample of 60 students attend a series of coaching classes before taking the test. Let μ be the population mean IQ score that would occur if every student took the coaching classes. The classes are successful if $\mu > 100$. A test is made of the hypotheses $H_0 : \mu \leq 100$ versus $H_1 : \mu > 100$.

- Assume that the classes are successful but the conclusion is reached that the classes might not be successful. Which type of error is this?
- Assume that the classes are not successful. Is it possible to make a Type I error? Explain.
- Assume that the classes are not successful. Is it possible to make a Type II error? Explain.